

Catching chip failures before end-of-line

A Snowflake-to-Tableau yield monitoring
pipeline, built on the UCI SECOM
semiconductor dataset.

Snowflake

Tableau Cloud

SPC + ML

No SQL for users

- WHAT IS A SEMICONDUCTOR

The material powering **every** modern device

Silicon wafers are sliced, etched, and layered into chips — the brains inside every phone, car, and server on earth.

smartphones

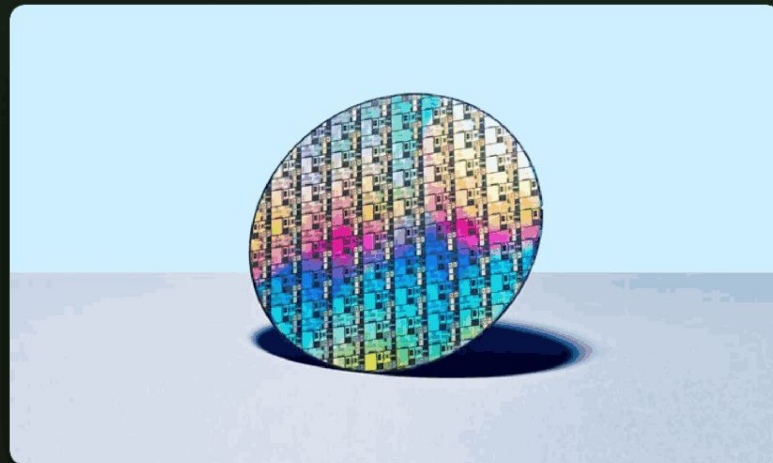
EVs

AI chips

medical devices

~\$600B

global semiconductor market



One bad sensor. Thousands of failed chips.

We built a pipeline to surface
failure signals before it's too late
— no SQL needed.

The problem

Failures only surface at end-of-line testing, after the cost is already sunk.

Our solution

Raw sensors → Snowflake → Tableau + ML, surfacing risk in near real time.

For whom

Process engineers and operations managers on the fab floor.

UCI SEC0M · REAL FAB DATA, 2008

• THE
DATASET

1,567 wafers. 590 sensors each.

1,567

production runs

590

sensor features

104

fails — only 6.6% failure
rate

Pass / Fail

binary yield label

- Heavy missing values → KNN imputation
- Zero-variance sensors removed in staging
- Class imbalance makes failure prediction hard



Clean, map, and serve SECOM data

01 · RAW

Load source

Preserve the uploaded table; original columns C1–C592, where C1 is the timestamp and C592 the label.

02 · STAGING

Clean & map

Remove the header row, cast timestamp and sensors, encode –1 → PASS and 1 → FAIL.

03 · MART

Build analytics

Daily yield & 7-day MA, failure drilldown table, SPC limits and outlier flags.

04 · SERVE

Tableau

Connect Tableau Cloud to curated MART tables — no SQL for the end user.

→ Keep transformation logic in Snowflake, then connect Tableau to curated MART tables for fast, governed dashboarding.

From a 592-column upload to **four curated tables**

DATASET 01

MART_YIELD_DAILY

Row-level test records rolled into one daily production summary — yield, pass/fail totals, 7-day MA.

DATASET 02

MART_FAIL_ANALYSIS

Filtered to failed units with time-drilldown fields — explains when failures happen.

DATASET 03

MART_SENSOR_STATS

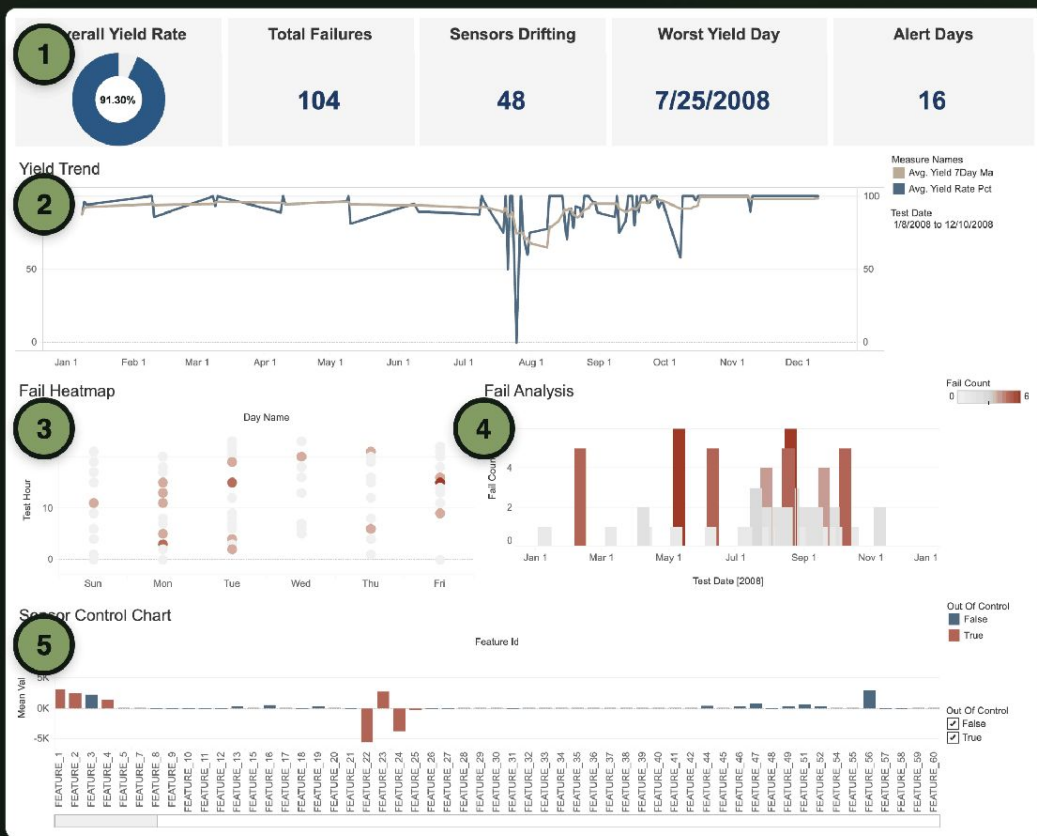
All 590 sensors summarized into one SPC reference — baseline mean, std dev, UCL, LCL.

DATASET 04

MART_SENSOR_CONTROL

Each sensor reading mapped to its control limits and status flag for the SPC dashboard.

The takeaway: the data is no longer a raw upload — it becomes business-readable datasets built for yield trends, failure drilldowns, and SPC monitoring.

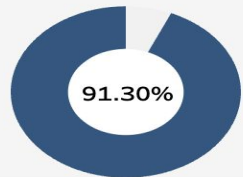


THE DASHBOARD

Five zones, one screen

- 1 Health KPIs**
Yield %, fails, drifting sensors, alert days — today at a glance.
- 2 Yield Trend**
Daily yield vs. 7-day average — spot dips before they spread.
- 3 Fail Heatmap**
When fails cluster — by weekday and hour of the shift.
- 4 Fail Analysis**
Which dates carried the failures — red bars are the bad batches.
- 5 Sensor Control Chart**
Per-sensor mean — red bars are out of control and need a look.

Overall Yield Rate



Total Failures

104

Sensors Drifting

48

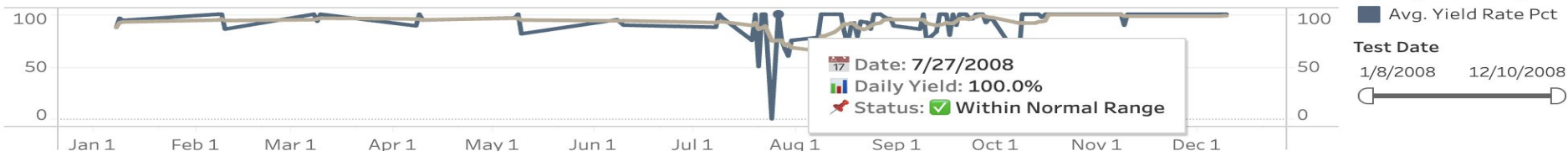
Worst Yield Day

7/25/2008

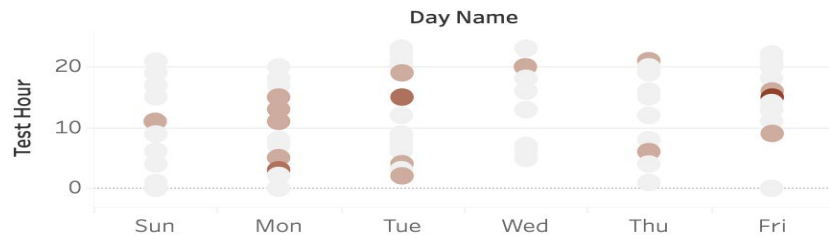
Alert Days

16

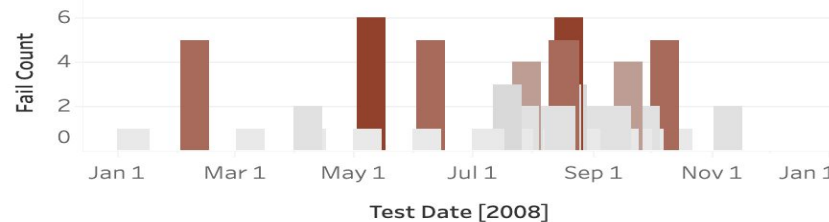
Yield Trend



Fail Heatmap



Fail Analysis



Sensor Control Chart





Welcome to the **SECOM Process Intelligence Dashboard**. I have deep knowledge of your **SECOM semiconductor manufacturing data**.

Production Summary — 2008

91.3%

Overall Yield

104

Total Failures

48

Sensors Drifting

16

Alert Days

Which sensors are drifting?

Worst yield day?

What caused the August failure?

Which shift has most failures?

AI Chatbot



Sensor Drift Analysis

Critical Drifters (Top 10% Std Dev):

- **FEATURE_22, FEATURE_23, FEATURE_24** – Primary culprits
- 48 total sensors showing out-of-control behavior
- Correlation: Drift onset ~late July 2008

Pattern:

- Systematic increase in variance during July–September anomaly window
- Coincides with 0% yield day (July 25) and 16 low-yield alert days
- Suggests equipment degradation or calibration drift

Actionable Steps:

1. **Immediate:** Pull SPC (Statistical Process Control) charts for FEATURE_22-24; check calibration logs for July 2008
2. **Root cause:** Equipment maintenance records → likely thermal/mechanical wear

Which sensors are drifting?



AI Chatbot Claude x Tableau Cloud

What is the yield trend in my Yield Trend view?

Used tableau integration, loaded tools >

Here's your **Yield Trend** view. The chart tracks two metrics over the year:

- **Avg. Yield Rate Pct** (dark blue line) — the daily yield rate
- **Avg. Yield 7-Day Moving Average** (tan/beige line) — a smoothed trend line

Key takeaways:

- **Jan-Jun:** Yield was relatively stable and healthy, mostly in the **90-100%** range with some moderate dips.
- **Late Jul - Aug:** A **severe disruption** — yield collapsed dramatically, with the daily rate hitting near **0%** at its worst. This is clearly a major quality or production incident.
- **Aug-Sep:** A **recovery period** with high volatility, bouncing between ~70-100%.
- **Oct-Dec:** Yield largely **recovered and stabilized** back near 100%, though there's a notable dip around late October (~59%).

The 7-day moving average smooths out the daily swings and clearly shows the trough bottoming out around early August before gradually climbing back. Would you like to dig deeper into what caused those failures — perhaps by looking at the **Fail Analysis** or **Fail Heatmap** views?



Write a message...



Sonnet 4.6 Low



Your morning routine, top to bottom

Read the board the way you'd walk the line — start at the headline numbers, then drill down until you know exactly which sensor to touch.

1 00:10

Is today okay?

Scan the **KPI strip**. Yield near 91%? Any new alert days? If normal, move on.

2 00:25

Where's the dip?

Follow the **Yield Trend**. When the daily line drops below the 7-day average, note the date.

3 00:45

Which batch?

Cross-check the **Heatmap & Fail Analysis**. Red marks the batch to pull.

4 00:60

Which sensor?

Drop to the **Sensor Control Chart**. Every red bar is where you intervene.

Thank you

Raw sensors in. A dashboard a process engineer can read in 60 seconds, out.

Snowflake

Tableau Cloud

UCI SECOM dataset